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CASE STUDY

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Towards responsible AI: an implementable blueprint for integrating explainability and social-cognitive frameworks in AI systems

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Abstract

Automation increasingly shapes modern society, requiring artificial intelligence (AI) systems to not only perform complex tasks but also provide clear, actionable explanations of their decisions, especially in high-stakes domains. However, most contemporary AI systems struggle to explain their runtime operations in specific instances, limiting their applicability in contexts demanding stringent outcome justification. Existing approaches have attempted to address this challenge but often fall short in terms of contextual relevance, human cognitive alignment, or scalability. This paper introduces System-of-Systems Machine Learning (SoS-ML) as a novel framework to advance explainable artificial intelligence (XAI) by addressing the limitations of current methods. Drawing from insights in philosophy, cognitive science, and social sciences, SoS-ML seeks to integrate human-like reasoning processes into AI, framing explanations as contextual inferences and justifications. The research demonstrates how SoS-ML addresses key challenges in XAI, such as enhancing explanation accuracy and aligning AI reasoning with human cognition. By leveraging a multi-agent, modular design, SoS-ML encourages collaboration among machine learning models, leading to more transparent, context-aware systems. The framework's ability to generalize across domains is demonstrated through experiments on the Pima Indian Diabetes dataset and pie chart image-to-text interpretation, showcasing its transformative potential in improving both model accuracy and explainability. The findings emphasize SoS-ML's role in advancing responsible AI, particularly in high-stakes environments where interpretability and social accountability are paramount.

Keywords Machine learning framework, Explainable artificial intelligence, Multi-agent systems, Social constructs, Cognitive sciences in AI, System of systems

Introduction

As artificial intelligence (AI) systems expand into high-stakes domains such as healthcare, finance, and autonomous systems, it has become increasingly clear that computational optimality often diverges from social optimality. This gap has sparked discussions around responsible AI, which emphasizes the need for systems that are both technically robust and aligned with social values. Traditional AI approaches prioritize model accuracy, but responsible AI calls for a balance between accuracy, transparency, fairness, and accountability. These elements are deeply interwoven, as neglecting responsible development can lead to deployment risks, such as user distrust, governance challenges, and ambiguity in accountability. For example, an autonomous wheelchair lacking safety overrides poses not only practical but also ethical challenges, highlighting the importance of context-awareness and generalizability.

A key component of responsible AI is explainability-the ability to understand and communicate how AI systems make decisions. However, current explainable AI (XAI)

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methods, such as LIME and SHAP [1, 2], often provide post-hoc explanations that are misaligned with human cognitive processes or the broader social context. These explanations, primarily aimed at developers, are often too technical for end-users, and they do not always accurately reflect the internal workings of the models, leading to concerns about reliability in critical applications such as healthcare and legal decision-making [3].

The current literature distinguishes between interpretability and explainability. Interpretability refers to understanding a model's internal mechanisms, typical of simpler models like decision trees or linear regression [3, 4], while explainability focuses on generating post-hoc justifications for predictions made by more complex models, like neural networks [5, 6]. This debate centers on whether post-hoc explanations are reliable enough for high-stakes applications. While some researchers advocate for inherently interpretable models [3], others argue that post-hoc methods are necessary for managing the complexity of advanced AI models [7, 8].

Moreover, while certain desirable attributes for XAI models-especially those that bridge the gap between social and computational sciences-have been identified in the literature [1, 2, 9–19], a significant question remains: how can we design AI systems that truly integrate these perspectives to provide explanations that extend beyond post-hoc justifications and effectively serve both developers and end-users? This suggests the need to redefine "explanation" in AI-not just as a means of achieving technical correctness through tools such as LIME or SHAP but as a broader, socially-driven concept. Such a shift would encourage explanations that function as social constructs, aligning more closely with human reasoning and interaction.

To address these gaps, we propose the System of Systems for Machine Learning (SoS-ML) blueprint, which extends beyond the traditional divide between interpretability and explainability by integrating both directly into the system's design. Built on principles derived from natural systems in biology and linguistic theories, SoS-ML emphasizes modularity, context-awareness, and functional interpretability. Instead of relying on post-hoc explanations, SoS-ML generates evidence-based explanations from the system's internal representations, ensuring transparency and reliability in decision-making. The modular structure also encourages the integration of multiple models and datasets through phased development, enhancing generalizability across diverse contexts.

The proof of concept for Phase 1 SoS-ML demonstrates its feasibility using the Pima Indian Diabetes dataset, where its multi-agent configuration improved accuracy by 5–10% over traditional models, achieving an overall accuracy of 80%, which is competitive with standard neural networks. Modular agents processed specific dataset subsets collaboratively, reducing uncertainty and improving prediction reliability. For instance, the group handling [pedigree, BMI, glucose] achieved a reliance score of 81% with a confidence interval of 75–87.5%, showing the framework's ability to target specific subtasks for refinement. No existing interpretable techniques were used when setting up the experiment.

A more sophisticated use of the proposed SoS-ML is demonstrated by the Pie Chart Interpreter, which illustrates SoS-ML's effectiveness in a more general context. The interpreter, designed to explain pie chart slice sizes, used multiple agents to analyze features such as color, intensity, and edges. The system's Q-learning agent dynamically selected the optimal combination of agents for interpretation, adjusting based on the complexity of the chart. For pie charts with ten slices, the combination of intensity and edge processing agents was selected 88.46% of the time, showcasing the system's ability to adapt contextually.

Thus, SoS-ML exemplifies a new paradigm in AI system design, integrating transparency and explainability from the outset. In addition to enhancing model accuracy and generalizability, the SoS-ML framework can also incorporate real-time human feedback, ensuring the system continuously adapts and refines its predictions based on evolving user input. This paper begins by detailing the current state of XAI, its strengths and shortcomings in "Related work" section. Then it covers the insights and AI development specifications we gather from natural and cognitive sciences in "Deriving architectural insights and specifications for developing a XAI system" section. Next, the paper proceeds to present a detailed overview of the SoS-ML blueprint in "Methods: defining the architectural blueprint that fits the derived specifications" section, proof of concept results and initial development outcomes in "Results and discussion" section, illustrating how the framework contributes to more responsible and effective AI deployment through the creation of error-checking mechanisms and explanation accuracy metrics.

Related work

Responsible AI and interpretability

The integration of interpretability within the framework of responsible AI has become increasingly important. Responsible AI emphasizes transparency, fairness, and accountability in machine learning systems. Doshi-Velez and Kim (2017) [5] and Dignum (2017) [20] argue that interpretability is central to building ethical AI systems, particularly in ensuring that models align with societal values and regulatory requirements. Molnar et al. [21] reinforces this by advocating for the need to make AI systems interpretable not just for technical users, but

for broader stakeholders including policymakers and end-users.

Distinction between interpretability and explainability

In the current literature, there is a distinction between interpretability and explainability that often gets blurred but is important to differentiate. Interpretability refers to the extent to which a model's internal mechanics can be understood by a human observer, meaning the model itself is transparent. Models such as decision trees or linear regression are considered interpretable because their structure and decision-making process are inherently simple and easy to follow [3, 4]. On the other hand, explainability refers to the ability to provide a post-hoc explanation for the predictions of a model, particularly for black-box models such as deep neural networks or ensemble methods. Explainability techniques, such as LIME and SHAP, do not necessarily make the model itself transparent but instead generate explanations for specific predictions [5, 6].

The debate between these two concepts centers on whether post-hoc explanations are reliable enough for high-stakes decisions, such as in healthcare or legal applications. Researchers such as Rudin [3] argue that inherently interpretable models should be preferred over black-box models, especially in scenarios where transparency and accountability are paramount, as post-hoc explanations can sometimes be incomplete or misleading. On the other hand, many practitioners rely on explainability techniques to make sense of complex, high-performance models, which may be necessary for tasks where accuracy significantly outweighs the need for full transparency [7, 8, 22].

In this paper, we aim to address the limitations posed by the traditional dichotomy between interpretability and explainability in AI. To reconcile these views and meet the demands of complex AI applications, we propose a shift in how AI systems are conceived. Rather than viewing interpretability and explainability as separate, post-hoc processes, we argue that AI models should be designed from the ground up to seamlessly integrate both.Our contribution introduces a new framework where AI systems are both modular and context-aware, providing interpretable insights while generating more reliable explanations throughout their lifecycle. We move from traditional "model interpretability", which focuses on individual components, to "functional interpretability", where transparency emerges from the modular design of system functions. Additionally, we shift from post-hoc explanations to evidence-based explanations derived directly from the model's internal representations, making explanations more integrated and robust. The SoS-ML framework introduced in this paper exemplifies this new approach by embedding transparency and explainability into the model's architecture, rather than treating it as an afterthought.

Exisitng XAI models

The need for interpretability in ML has grown as AI systems are applied to high-stakes areas such as healthcare, finance, and autonomous systems. The complexity of advanced ML models, particularly deep learning and ensemble methods, often leads to a trade-off between accuracy and transparency. This has prompted a growing body of work dedicated to XAI [21], focusing on developing tools and methodologies for understanding how complex models make decisions.

Global interpretation

Global interpretation techniques are designed to explain the overall behavior of a model, offering insights into how various features contribute to predictions across the entire dataset. These methods help identify trends and interactions in model decision-making, and are widely used in many machine learning applications.

- **Partial Dependence Plots (PDPs)**: PDPs are one of the most commonly used tools for global interpretation. They show the marginal effect of one or more features on the predicted outcome. PDPs have been applied across various domains, including clinical decision support systems and predictive maintenance. However, they can suffer from misleading interpretations in the presence of correlated features.
- **Accumulated Local Effects (ALE) Plots**: ALE plots, proposed by Apley & Zhu (2020) [23], improve upon PDPs by addressing the issue of correlated features. ALE plots partition feature values into intervals and average their effect, thus offering a more accurate representation of feature influence. These plots have been widely adopted in industrial applications and fields such as climate science.
- **Permutation Feature Importance**: Introduced by Breiman (2001) for random forests [24, 25], permutation importance has since been applied broadly across various models. By shuffling feature values and measuring the change in model accuracy, this technique provides a robust, model-agnostic measure of feature relevance [26, 27]. It is particularly useful in finance and healthcare for ranking features in credit scoring and risk prediction systems [28, 29].

Local interpretation

While global methods provide broad insights, local interpretation focuses on explaining individual predictions.

This approach is valuable in high-stakes applications where specific decisions need to be justified or contested, such as loan approvals or medical diagnoses.

- **LIME (Local Interpretable Model-Agnostic Explanations)**: LIME, developed by Ribeiro et al. (2016) [1], has become a widely adopted tool for generating explanations for individual predictions by perturbing inputs and fitting an interpretable surrogate model locally.
- **SHAP (SHapley Additive Explanations)**: SHAP, based on cooperative game theory, provides consistent, theoretically grounded explanations by assigning each feature a contribution value for each prediction. Lundberg and Lee (2017) [29] developed SHAP, which has been used extensively [15, 16] in fields such as healthcare for explaining complex models in patient diagnosis systems. SHAP's ability to provide both local and global insights has made it a versatile tool across domains.
- **Counterfactual Explanations**: Counterfactual methods aim to show how a model's prediction would change if certain input features were altered. Wachter et al. (2017) [30] introduced this technique, which is particularly useful in fields such as law and finance, where explainability is tied to actionable insights (e.g., "If your income were higher, your loan would be approved"). Counterfactuals are seen as intuitive and user-friendly because they align with how humans often think about decision-making [31].

Model-agnostic methods

Model-agnostic methods are critical because they provide interpretability for any black-box model, regardless of its architecture. This versatility makes them particularly important in real-world applications where a variety of models may be used. Examples include LIME, SHAP, and permutation feature importance, among many others.

Inherently interpretable models

While post-hoc explanations provide a means to interpret black-box models, inherently interpretable models offer transparency from the outset. These models are simpler by design and include approaches such as:

- **Generalized Additive Models (GAMs)**: GAMs, popularized by Hastie and Tibshirani (1986) [32], model each feature separately, offering a balance between interpretability and flexibility. They have been successfully applied in epidemiology and energy consumption prediction.
- **Decision Trees**: Decision trees [33] remain one of the most interpretable models due to their clear, rule-based structure. They are widely used in healthcare, credit scoring, and other domains where transparency is essential.

Challenges and shortcomings of existing XAI

Several real-world challenges of XAI systems highlight the difficulty of balancing accuracy and interpretability. Here are a few notable examples:

- Dressel and Farid [34] COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is an algorithm used in the U.S. criminal justice system to predict the likelihood of a defendant reoffending. Despite efforts to provide transparency, COMPAS faced significant criticism for racial bias in its predictions, disproportionately labeling Black defendants as high-risk compared to White defendants. Explanations provided by COMPAS were insufficient to address these biases, raising ethical concerns about using opaque algorithms in sensitive areas such as criminal justice. This failure illustrates that post-hoc explanations do not necessarily ensure fairness, and that interpretability does not always guarantee responsible AI.
- Barlas et al. [35] In 2015, Google's image recognition system, powered by machine learning, mistakenly tagged African Americans as "gorillas". The failure highlighted the shortcomings of explainability techniques in image recognition systems. While the system was highly accurate overall, this catastrophic error was the result of biased training data. Google's system lacked effective explainability mechanisms to ensure that biases in the training process were detectable and correctable before deployment, resulting in severe public backlash.
- Banks et al. [36] Tesla's Autopilot system, which uses AI for autonomous driving, has been involved in multiple accidents. In some cases, the system failed to recognize obstacles, such as trucks or pedestrians, leading to fatal crashes. One notable incident occurred in 2016 when the system failed to distinguish between a bright sky and a white truck, resulting in a fatal collision. While Tesla has used machine learning models to power its Autopilot system, these models' decision-making processes remain largely opaque, and explanations provided by Tesla post-incident have not addressed the fundamental issue of why the system failed to recognize certain objects.
- Frank [37] IBM's Watson for Oncology was once touted as a game-changer for cancer treatment rec-

ommendations, but it failed to live up to its promise. Watson was supposed to assist doctors by analyzing large datasets and providing treatment suggestions. However, the system made several incorrect or unsafe recommendations based on limited and biased training data. Explanations provided by the system did not adequately account for the limitations in data quality, leading to distrust among healthcare providers. This failure highlights how incomplete explanations in critical domains such as healthcare can result in a loss of confidence and effectiveness.

- New York State Department of Financial Services [38] Apple's credit card, managed by an AI algorithm, came under fire in 2019 after users noticed that women were being given significantly lower credit limits than men, despite having similar or better credit histories. While the company claimed the system was unbiased, the lack of transparency and insufficient explanations for the gender discrepancies made it difficult for users to understand the decision-making process. The failure of Apple to provide clear, explainable reasoning behind credit limit decisions led to public criticism and regulatory investigations.

SoS-ML addresses these issues or at least reduce the adverse impact by integrating **context-aware** explanations and **functional interpretability**, ensuring that the model not only provides transparent, evidence-based explanations but also allows for the detection and mitigation of biases within its components.

Deriving architectural insights and specifications for developing a XAI system

Figure 1 illustrates the sequential stages in the development, deployment, and management of AI systems, grounded in insights from both social and natural sciences. The process begins with an interdisciplinary foundation ("Social/Natural Science"), which informs two primary pathways: the "Developmental Pathway" and the "Deployment Pathway". The former directs the system through "AI Development and Testing", ensuring that the AI system is rigorously developed and evaluated according to specific characteristics or specifications before deployment. The latter pathway guides the system through the "AI Deployment" phase, followed by "AI Management", which oversees the operational lifecycle of the AI system. This flowchart emphasizes the crucial role of interdisciplinary knowledge in shaping AI development and ensuring robust and responsible deployment and management.

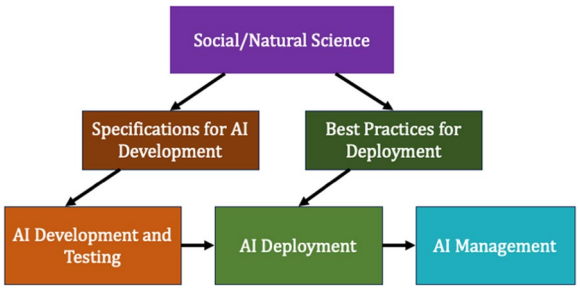


Fig. 1 Integration of social and natural sciences into AI system development and other stages of its life cycle

Two key questions emerge: which principles from human-centered sciences should be used to guide AI development, and how these insights can inform the design and specifications of XAI systems-not just at the theoretical level, but also in the practical developmental implementation of AI systems.

Adopting a human-centered science perspective. Intelligence and explanation have long been studied in both the natural and social sciences.

1. Moreover, the origins of state-of-the-art AI systems can be traced to the development of neural networks, and subsequently deep neural networks, which were inspired by the synaptic connections in the human brain. Thus, the human brain provides a natural starting point for developing AI systems.

2. Furthermore, since the goal is for the outputs of AI or XAI models to produce explanations that resemble the social constructs e.g. narratives used in human conversation, it is essential to consider theories that explain the human brain's ability to engage in dialogue. The two leading theories in this domain are Chomsky's Innatist Theory and Vygotsky's Social Interactionist Theory.

3. Additionally, to ensure that AI outputs conform to the expectations of narrative explanation, it is important to understand how explanations are interpreted in cognitive communities. This understanding will help create explanations that are intuitive for AI users and useful for developers. Aligning both needs allows AI systems to provide explanations that are accessible to users while offering the technical depth developers need for analysis and improvement.

In this section, we briefly discuss the relevance of human brain function and linguistic theories to the current topic. A comprehensive discussion of these topics is beyond the scope of this paper. We then present the derived specifications from these insights. The blueprint and proof of concept will be discussed in the subsequent sections ("Methods: defining the architectural blueprint that fits the derived specifications" and "Results and discussion" sections, respectively).

Derived insights from human-centered approach

Insights From the Human Brain. Insights from the Human Brain. Leading AI systems, particularly deep learning models, are often referred to as "black boxes" due to their lack of transparency, making it difficult even for developers to understand how decisions are made. This opacity arises from computational complexities such as iterative backpropagation, which hinders the tracing of individual component contributions and diffuses accountability. Similarly, the human brain operates as a "black box", where no single neuron is solely responsible for a specific function. Instead, brain functions emerge from the coordinated activity of various regions or networks of neurons.

Neuroscientists study the brain through *regions of interest (ROIs)*-clusters of neurons linked to specific functions. For example, language processing [39] involves multiple ROIs, such as Broca's area (associated with speech and grammar) and Wernicke's area (responsible for language comprehension). These regions work together to enable complex cognitive functions, as seen in the coordination of the brain's language centers (Fig. 2). Inspired by this, we propose **functional XAI (fXAI)**.

Definition 1 (fXAI System). We define fXAI as AI systems where developers may not understand the contribution of an individual neuron or layer, but they do understand what groups of these components contribute to the overall decisions, similar to ROIs of the human brain. This understanding may be translated to explanations, ideally, by the AI system itself.

Insights from Linguistic Theories: Chomsky and Vygotsky. Two key linguistic theories further inform the development of explainable AI:

Chomsky. Noam Chomsky's Innatist Theory [39, 41] posits that certain aspects of language acquisition are innate and biologically determined. Chomsky introduced the concept of a *Language Acquisition Device (LAD)*, a hypothetical mechanism that enables humans to naturally acquire language.

Applying this idea to AI, systems could be designed with built-in communication and explanation capabilities, similar to how the LAD operates in human language development. This suggests that AI architectures should incorporate intrinsic modularity, where specific components-referred to as **functional units**-are designed to handle distinct, explainable functions from the outset, rather than relying solely on learning these abilities. These **functional units** would also serve as the AI equivalent of regions of interest (ROIs) in the human brain, responsible for specific, understandable tasks within the system.

Vygotsky. The Social Interactionist Theory [42–46] emphasizes that cognitive development, language learning, and explanation capabilities are shaped by social interactions. Two key concepts relevant to AI systems are: i) Zone of Proximal Development (ZPD): Refers to tasks that learners cannot complete alone but can accomplish with guidance, and ii) Scaffolding: Involves providing temporary support within the ZPD to help learners progress.

Applied to fXAI, Vygotsky's theory emphasizes the importance of information exchange or **interaction** between functional units. It also highlights the need for diverse, hierarchical units to collaborate in a supportive environment to enhance task performance. This requires

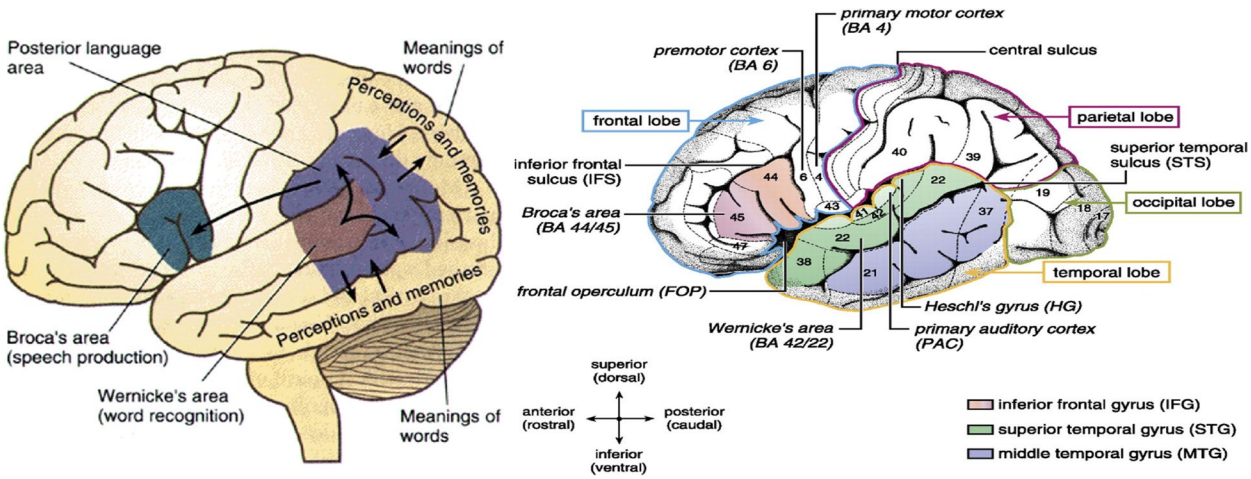


Fig. 2 The Language Center in the human brain. Reproduced from [40]

multiple units with both similar and distinct roles, working within a structured hierarchy to achieve the desired output.

Insights From Cognitive Science. Explanations, based on findings from philosophy, cognitive science, and psychology [41, 47–52], are fundamentally contextual social constructs that transfer knowledge through conversations or interactions. They exhibit the following characteristics:

- Contrastive Nature: Explanations are often contrastive [53, 54], answering why one event occurred instead of another (e.g., why event P happened instead of Q).
- Biased Selection: People tend to focus on a limited number of contextual causes, influenced by cognitive biases, rather than seeking complete explanations [47, 51]. This contrasts with computational explanations, where probability plays a significant role and causality is often regarded as the highest standard for quality explanations.
- Limited Role of Probabilities: While truth and likelihood matter, causal explanations are often preferred over statistical probabilities, as the most probable explanation may not be the most satisfying [55].

These characteristics suggest that user-facing AI explanations need to be contrastive, focusing on the most relevant causes within the context. This implies that the accuracy of AI-produced explanations should be evaluated based on how well they identify the best evidence from the available information. Grice [56] and Lombrozo [57] support this idea, with Grice viewing conversations as information transfer and Lombrozo arguing that humans prefer explanations that support the best inference rather than absolute causality.

Therefore, we propose that **AI contextual explanations** are structured as an evidence-inference pair:

$$Contextual\ Explanation_{AI} = Context(Inference + Evidence), \quad (1)$$

where, $Contextual\ Explanation_{AI}$ is a proper subset of social explanations and overlaps with causal explanations, indicating that AI explanations should be influenced both by social and causal reasoning (Fig. 3).

Implementation Implications of These Human-Centered Insights. In developing fXAI, it is crucial to structure the AI architecture around the functional units, which mirror the organized, modular functionality necessary for cognition (i.e., context-awareness in AI). Since these functional units should not be involved in the backpropagation process (see Fig. 4), they must operate independently while still interacting to perform and accomplish complex predictive tasks. This modular approach ensures transparency, as the function of each unit is explainable, even if the computations (or unit itself) remain complex.

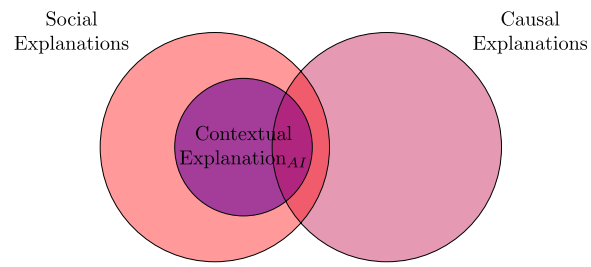


Fig. 3 Venn diagram illustrating the relationship between Contextual Explanations in AI, Social Explanations, and Causal Explanations. Contextual Explanation$_{AI}$ is a proper subset of Social Explanations and overlaps with Causal Explanations, indicating that AI explanations are both informed by social constructs and influenced by causal reasoning

Translating the insights into specifications

Based on the insights obtained, AI systems should be developed with an interdisciplinary foundation, integrating insights from both social and natural sciences, and be guided by human-centered design principles. These systems should draw inspiration from the modular structure of the human brain and linguistic theories, employing specialized, transparent functional units that collaborate in a structured environment. Additionally, AI explanations should be contextual, blending inference and evidence, and designed to be both intuitive for users and technically detailed for developers.

To effectively structure the development process, we propose dividing the specifications into three developmental stages: **Phase 1**, **Phase 2**, and **Phase 3**. By progressing through these phases, the system evolves from foundational agent and environment design (Phase 1), to more advanced interaction and communication (Phase 2), and ultimately to full-scale collaboration and control (Phase 3). In "Results and discussion" section, we demonstrate that each phase results in a fully functional, self-contained system, with its ability to handle complex tasks significantly increasing as we progress through the phases. This phased approach ensures that AI systems remain transparent, robust, and aligned with human-centered goals (see "Does SoS-ML conform to responsible AI?" section). The specifications for each phase are as follows:

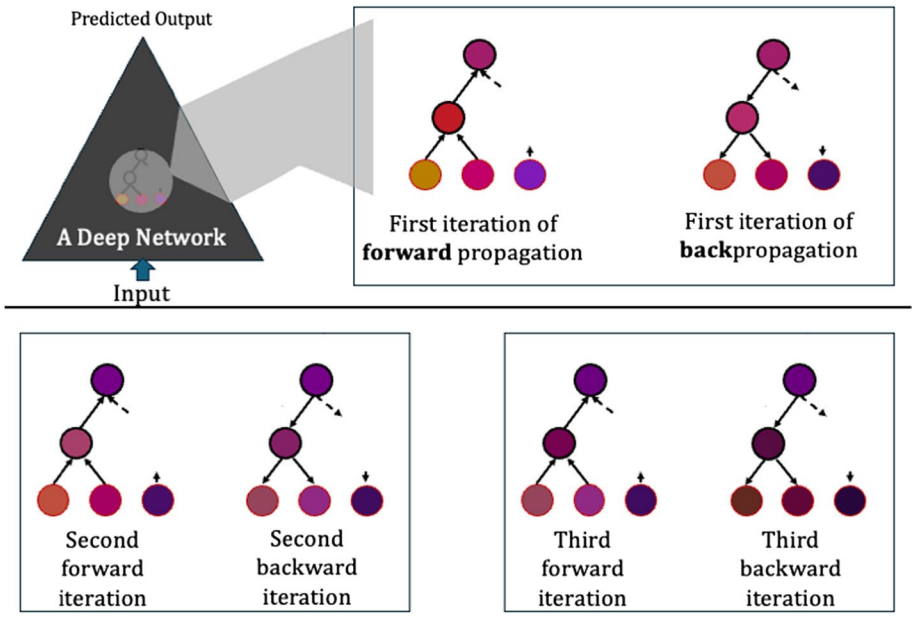


Fig. 4 This illustrates how backpropagation disperses the contribution of each neuron across the network, making it increasingly difficult to trace individual impacts, thereby turning the model progressively more opaque with each iteration

- **Environment Design**:
  - **Phase 1:** Establish a context-rich environment where agents can derive explanations and evaluate the accuracy of decisions.
  - **Phase 2:** Foster an environment that supports multiple interacting AI agents, promoting diversity and a sense of belonging.
- **Functional Unit Characteristics**:
  - **Phase 1:** Define a diverse group of functional units, referred to as **Data Agents**, responsible for processing data. These will act as the system's regions of interest (ROIs).
  - **Phase 1:** Develop functional units called **Interaction Platforms** that enable seamless information exchange between Data Agents. These platforms generate the inference-evidence pair and function as the system's primary Language Acquisition Device (LAD).
  - **Phase 2:** Create **System Agents** to coordinate system-level and contextual information. These agents can serve roles akin to LAD or Zone of Proximal Development (ZPD).
  - **Phase 2:** If necessary, introduce functional units, called **Information Channels** that allow for straightforward information transfer, without any processing between various functional units.
  - **Phase 2:** Specialize certain functional units for communication with humans, particularly end users. These units will transform raw inference, evidence, and contextual information into Contextual Explanation$_{AI}$.
  - **Phase 2:** Define **Concept Agents**, designed to incorporate human feedback and domain knowledge, functioning as scaffolding to approximate human understanding.
- **Extended Collaborative AI System Goals**:
  - **Phase 3:** Expand the number and diversity of functional units to enhance system capabilities such as processing multi-modal data.
  - **Phase 3:** Integrate diverse information and communication pathways to combine justifications for decisions, including hypothetical and testable facts.
  - **Phase 3:** Implement control mechanisms to identify and report weaknesses and biases within the interactive system.

Methods: defining the architectural blueprint that fits the derived specifications

In this paper, we propose that the AI system most conducive to explainable AI (XAI) and social explanation is a System of Systems for Machine Learning (SoS-ML). The SoS-ML framework is designed to meet the derived

specifications through a modular, multi-agent architecture that fosters transparency, collaboration, and contextual explanations across various phases of development.

Before defining the concept of an SoS and detailing the proposed architectural blueprint, we first present four scenarios that illustrate how the SoS-ML behaves at different stages of development. These scenarios demonstrate the system's behavior in Phase 1 (1 scenario), Phase 2 (1 scenario), and Phase 3 (2 scenarios), showing how the AI system evolves to handle increasingly complex tasks. By showcasing these scenarios, we aim to highlight the practical applicability and generalizability of the blueprint across multiple AI applications.

Scenario 1, Phase 1: Multi-Agent Configuration. Suppose an AI system's task is to predict the development of diabetes based on predictor variables such as the number of pregnancies the patient has had, their BMI, insulin level, age, and so on. This AI system consists of three colonies. The colonies are exactly the same except that they have access to slightly different data view e.g. only one colony can see the original dataset, while the other two can only access bootstrapped [CITE] versions of the dataset. Each colony consists of multiple agents. Each agent within a colony has access to only a pair of variables, and each slightly different from the next either in hyperparameter or prediction algorithm. When the AI system makes a prediction, the agent within each colony first undergoes interaction via weighted majority voting, followed by similar interaction among the colonies. Consensus among the colonies is taken as strong support for the correctness of the prediction. Prediction variations among the agents with same algorithm, hyperparameters, and variable access across all colonies is used to construct confidence intervals as an estimate of variable pair contribution to the prediction made by the AI system.

Scenario 2, Phase 2: Multi-Agent Configuration with System Control and Limited Contextual Interaction. Imagine an AI system designed to figure out and explain the sizes of slices in a pie chart just by looking at the image. To process the image, the AI system consists of three agents: Agent 1 focuses on color, Agent 2 on intensity, and Agent 3 on edges. They come together to interact and predict the sizes of the slices. However, to provide explanation, the AI system needs to understand how the slice sizes relate to one another (e.g. 10% < 90%), a very human notion. Thus, the AI system uses Agent 4 as the mechanism to bridge this communication gap. Agent 4 acts as a communication bridge within the system, synthesizing the data from Agents 1, 2, and 3 to form a coherent analysis of how the slices compare to each other in size by contextualizing the data in human-understandable terms, such as percentages and their comparative significance. Agent 5, a variant of a large language model, takes the outputs from Agent 4 and translates the technical data into natural language descriptions. This makes the information accessible and easily understandable to users, effectively explaining the composition of the pie chart in plain English. In order to provide evidence-based inferences, the AI system comes with another agent, Agent 6, which allows the AI system to methodically turn off certain functions and arrive at a conclusion that even though intensity is the most robust factor (among color, edge and intensity) for accurate slice size prediction, the importance of edge functionality increases as the complexity (aka. number of slices) of pie charts increases.

Scenario 3, Phase 3: Multi-Agent Configuration with Advanced System Control and Full Contextual Response. Suppose an AI system's task is to predict employee salary based on their age, education level, job title, and years of experience. Suppose this AI system initially consists of: Agent 1, who is an expert trained on data from all past and presents employees in the company; and 2 System Agents with an environment that enhances the system's capability of context detection and response. While processing the data, the System Agents trigger a check for gender bias in the dataset. This results in the creation of two more agents: Agent 2 specializes only on prediction for female employees; and Agent 3 specializes only on male employees. During an interaction between Agents 2 and 3, which involves comparing employee profiles, they find that the same employee profile has a lower score under the Agent 2's prediction than under Agent 3's prediction. They interact and consult with Agent 1 and find that Agent 3's prediction is supported more by Agent 1's prediction for the same employee. With this evidence, Agent 2 decides to issue a caution notification for gender bias. On interaction with a human expert, Agent 2 and Agent 3 both provides testament that given all other factors being equal, years of experience earns more salary for a male employee than a female employee.

Scenario 4, Phase 3: Fully Developed SoS-ML. Consider an AI system designed to assist clinicians in detecting and managing hypertension. This digital assistant is capable of providing detailed insights about hypertension, analyzing data patterns, recommending medication adjustments, offering personalized lifestyle suggestions with reasoning, and utilizing an alert system to notify both patients and clinicians of persistent irregularities while minimizing false-alarm fatigue. The AI system, modeled as a SoS-ML (System of Systems for Machine Learning), is displayed in Fig. 5 and organized as follows:

- System A consists of four ML models trained on different samples of lifestyle datasets. These models interact, identify consensus or disagreement, and produce logically consistent, evidence-based infer-

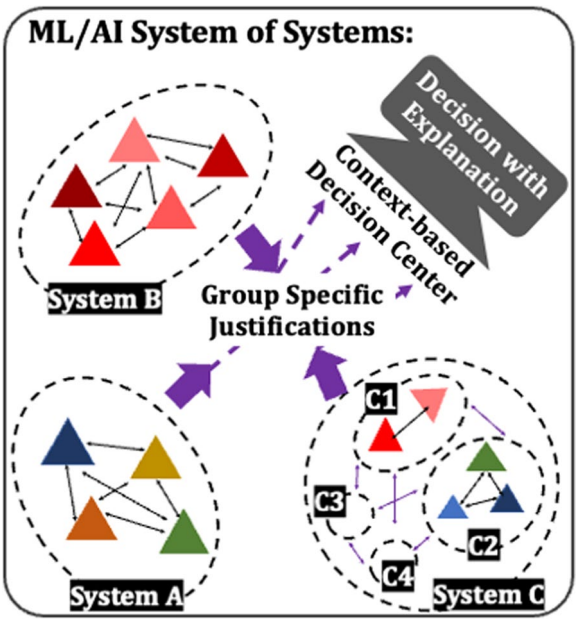


Fig. 5 An schematic example of a fully developed SoS-ML

ences. By training different models on various data samples, the system captures diverse perspectives on the same lifestyle concept.

- System B includes five ML models trained on gene expression data, each specializing in classification tasks using a maximum of ten genes. This system focuses on leveraging genetic information for hypertension management.
- System C is a complex network of interconnected systems trained on electronic medical record (EMR) data. Subsystems like C1, C2, and others are trained on specific EMR tables (e.g., encounters, medication, lab results). Notably, System C2 includes three ML models trained separately on female-only, male-only, and combined gender data to detect potential gender biases.
- System D, or the Decision Center, comprises three models: one combines the reports from Systems A, B, and C; another learns over time to optimize decisions by choosing between models; and a third validates the SoS-ML's output against a corpus of domain knowledge. Optionally, the output can be translated into human language using a large language model (LLM) for improved user understanding.

System of systems, SoS

A **System of Systems (SoS)** is a systems engineering framework [58, 59], where a **system** consists of interconnected components working together to achieve a common goal. The components exhibit interdependence, creating behaviors and properties not found in the individual elements. For example, a car's components-engine, transmission, wheels- work together for transportation, a function that no single part can achieve on its own.

In contrast, a **System of Systems (SoS)** is a collection of independent systems that collaborate to achieve a higher-level goal. Each system retains autonomy and its own functionality, but their collaboration produces emergent properties beyond the sum of the individual systems. For example, an SoS could include cars, traffic systems, and public transport networks working together to improve overall mobility, safety, and efficiency. Each system maintains its autonomy while contributing to the shared goal.

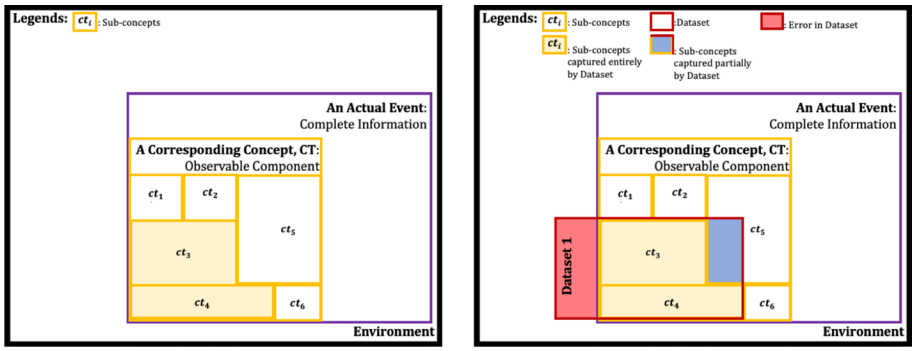
Five key elements differentiate an SoS from a traditional system:

- **Autonomy:** In a system, parts lose individual autonomy to operate under centralized control. In an SoS, each system retains autonomy and self-governance while contributing to the SoS's goals.
- **Connectivity:** In a system, connectivity is predefined and limited. In an SoS, connectivity is dynamic, allowing systems to create flexible, networked connections to enhance overall capability.
- **Diversity:** A system minimizes diversity through modular design. An SoS embraces diversity, leveraging the unique capabilities of each system to adapt to uncertainty and foster innovation.
- **Belonging:** In a system, parts inherently belong to the system. In an SoS, systems choose to participate based on the benefits they gain from contributing to the larger purpose.
- **Emergence:** In a system, emergence is planned and controlled. In an SoS, emergence is unpredictable, allowing for adaptability, innovation, and early detection of undesirable behaviors.

Phase 2 AI systems described in "Translating the insights into specifications" section of this paper embody autonomy, connectivity, and diversity. Belonging and emergence are expected to be fully realized and tested in Phase 3 systems.

Proposed blueprint: the abstract SoS-ML

In defining the abstract SoS-ML, we use the idea of *information* and set theory notations (Fig. 6). Broadly, there are three kinds of fXAI functional units within these blueprints: agents, communication mechanisms, and support devices.



(a) The space of incomplete but pure information, the world and a concept.

(b) Dataset introducing error due to limitations encountered by human agents.

Fig. 6 The basic components of the Environment

The environment

Imagine an environment containing all information about everything. This space, referred to as the **environment**, includes **actual events**, which are stand-alone occurrences or collections of interconnected events. In this environment, all information is **complete**, with no hidden variables.

However, agents-whether human or ML/AI-only have access to **incomplete information** about actual events. The portion of the environment accessible to an agent is called the **world**, and a concept within this world is denoted as ${}^{id}CT$, made up of smaller **sub-concepts**, ${}^{id}ct_i$. Formally:

$$ {}^{id}CT \subset W_{in},\ and\ {}^{id}CT = \bigcup_{i=C} {}^{id}ct_i,\ and\ ct_i \in \mathbb{R}^n \tag{2} $$

For example, if W_{in} represents all information on diseases, ${}^{HyperT}CT$ would represent all knowledge about hypertension, with ct_1 as *age*, ct_2 as *gender*, and so on. Incomplete information leads to sub-concepts, such as ct_5, combining the effects of factors like *life stress* and *economic status*.

In reality, human agents collect data to approximate ${}^{id}CT$, but datasets introduce errors. A dataset, D_j, may contain complete sub-concepts (ct_i), partial sub-concepts (${}_f ct_i$), and errors (e_j). Thus:

$$ {}^{id}D_j = \left(\bigcup_{i \in \mathbb{N},\ 0 \le f \le 1} {}_f ct_i \right) \bigcup (e_j) \tag{3} $$

To improve the accuracy of datasets in approximating ${}^{id}CT$, they must include repeated measurements, samples from diverse populations, and multiple data modalities. An approximation of ${}^{id}CT$ would be represented as:

$$ {}^{id}\widehat{CT} = {}^{id}D_{rm,ds,dp,dm} \tag{4} $$

$$ {}^{id}D_{rm,ds,dp,dm} = \left(\bigcup_{r=1}^{R} {}^{id}D_{rm} \right) \bigcup \left(\bigcup_{s=1}^{S} {}^{id}D_{ds} \right) \bigcup \left(\bigcup_{p=1}^{P} {}^{id}D_{dp} \right) \bigcup \left(\bigcup_{m=1}^{M} {}^{id}D_{dm} \right) \tag{5} $$

Here, R, S, P, and M represent repeated measurements, samples, populations, and data modalities, respectively, which help reduce errors and expand the knowledge available to agents.

The two additional key aspects of the environment-the interaction platform and selection pressure-are discussed in "Agent-agent interaction" section, following the introduction of AI agents in "The agents" section.

The agents

Within the proposed blueprint, there are three kinds of agents: i) *Data Agents* (e.g., ML models that process datasets), ii) *Concept Agents* (e.g., ML models or scripts that approximate human understanding of concepts), and *System Agents* (e.g., ML models that organize system and contextual information).

Defining an ML-based AI Agent Using Aristotle's Four Causes. In defining an ML-based AI agent, we propose four key components based on Aristotle's Four Causes:

1. **Material Cause**: This refers to the structural elements (e.g., neurons, support vectors, weights) and data components (datasets and features) that form the basis of the AI agent.
2. **Mechanism Cause**: The processes by which the agent learns and processes information, such as backpropagation, stochastic gradient descent, contrastive learning, and kernel approximation.

3. **Essential Properties**: These include the agent's cognitive abilities (e.g., counting, grouping, extrapolation), physical attributes (e.g., mobility), and technical specifications (e.g., hyperparameters, processing time).
4. **Final Product Cause**: The tangible outputs generated by the agent, such as class labels, text, or categories, which represent the agent's performance and utility.

Training Requirements for Independent AI Agents. To become independent, agents must undergo comprehensive ML training, which covers their structure, learning mechanisms, essential properties, and measurable outcomes.

Internal Representation and its Impact on Inference and Evidence: An agent's internal representation-the way it organizes and processes knowledge-defines its understanding and limits. This representation is shaped by the material, mechanism, and essential properties, affecting how well the agent can infer and provide evidence. The unique nature of each agent's perspective introduces subjectivity, making it crucial to utilize multiple agents to optimize the overall available information.

Agent Complexity and its Role in Explanation Quality: An agent's complexity, both mathematical and computational, forms the basis of its explanatory capacity. While complexity may not directly reveal insights into an agent's behavior, it plays a critical role in understanding why certain patterns or decisions emerge. For example- if the dataset demonstrates non-linear pattern, then the internal representation of a linear agent will not have good enough overlap with the dataset, resulting poor quality evidence. Metrics like the VC dimension help gauge this complexity, offering a foundation for evaluating the agent's internal representation and performance.

Components of Internal Representation. As shown in Fig. 7, an agent's internal representation is composed of several key elements:

- The part of the dataset corresponding to the concept (${}^{id}CT$), denoted by ${}^{id}_{f}D_j$.
- Inferred or generalized aspects of the concept not directly in the dataset, represented as ${}^{id}_{g}CT$.
- Errors from the dataset (te_j), as well as biases from the agent itself ($e\ bias$).

$$IR_{Agt} = \left({}^{id}_{f}D_j\right) \bigcup \left({}^{id}_{g}CT\right) \bigcup \left(e_{bias} \cup {}_{t}e_j\right), \quad (6)$$

where for all practical purposes $g \ll f$.

Final Conversion Loss. While the final product of an agent is derived from its internal representation, they are not identical. For example, in binary classification,

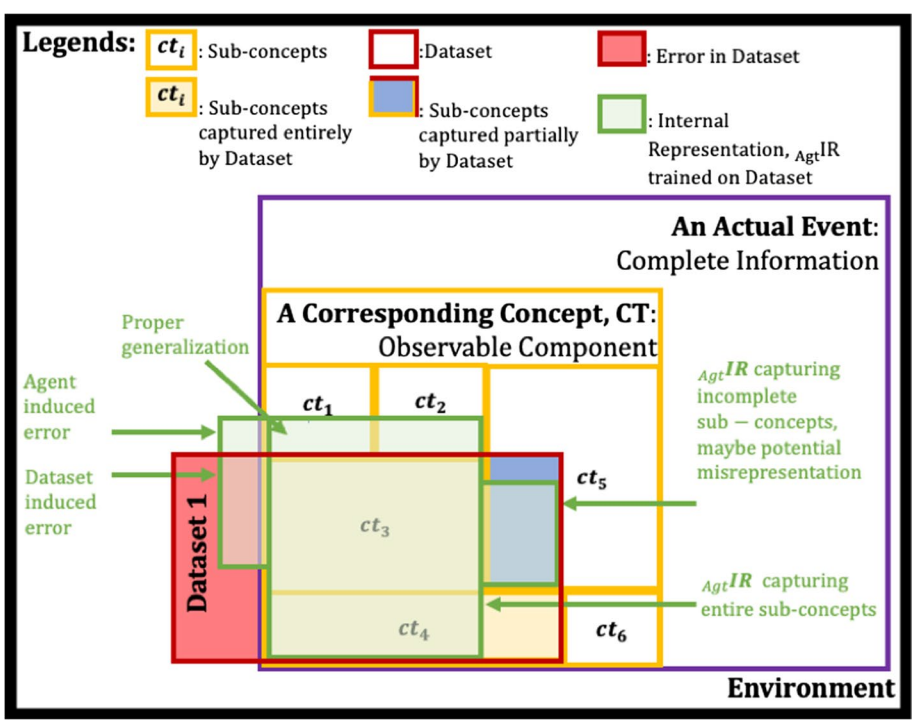


Fig. 7 The internal representation of an AI agent, when represented as a set of information

the final output format (e.g., 0,1 or −1,1) does not fully reflect the complexity of the internal representation. This information loss, termed **Final Conversion Loss**, occurs when converting the internal representation to the final output, as shown by the function h:

$$Final\ Conversion\ Loss = IR_{Agt} - \hat{IR}_{Agt}. \tag{7}$$

Acknowledging this loss highlights that the final output may not perfectly preserve the information encoded in the internal representation, potentially introducing biases or distortions during the conversion process. In multi-agent systems like SoS-ML, this loss can have significant implications when agents collaborate.

Agent-agent interaction

Agent-agent interaction is crucial for SoS-ML, as it enables AI models to collaborate, enhance their understanding, and improve decision-making. This interaction allows models to share information, adapt to complex situations, and collectively contribute to better, more robust outcomes.

Definition 2 (Definition of Agent-Agent Interaction). Consider two AI agents, Agt_i and Agt_j, with their respective internal representations, $IR_{Agt,i}$ and $IR_{Agt,j}$, trained within the same concept ${}^{id}CT$. Interaction occurs when one agent reveals its internal representation to the other, and the shared information is combined and resolved by both agents. This can be one-way (only one agent reveals) or two-way (both agents reveal their internal states).

Types of Interaction. Understanding the type of interaction required when developing an AI system using the proposed SoS-ML blueprint can guide the effective implementation of the LAD (Fig. 8b) and the Interaction Platform. Interactions between agents in a SoS-ML can be categorized based on the information processed, the directionality of communication, and the types of agents and datasets involved.

In terms of information exchange, interactions can either be passive or active (Fig. 8a). In *passive interactions*, agents share their information without altering their internal representations. The final outcome is simply a combination of their outputs, allowing for collaboration without requiring internal adaptation. In contrast, *active interactions*, which typically occur in Phase 3 systems, involve one or more agents retraining themselves based on the interaction. This process creates updated internal representations, reflecting the agent's adaptation to new information from its peers.

The directionality of interactions further classifies them into dominant, interdependent, or creative. In *dominant interactions*, one agent's output exerts significant influence, dominating the combined result. This can lead to an imbalanced outcome, where the perspective of the dominant agent overshadows others. On the other hand, *interdependent interactions* involve mutual learning, with both agents contributing equally to the final product, resulting in a balanced and collaborative outcome. *Creative interactions* take this a step further, as both agents work together to generate novel information, leading to results that could not have been predicted from their individual outputs alone.

Additionally, interactions can be classified based on the type of agents and the datasets they use. When two agents belong to the same machine learning algorithm family but differ in hyperparameters, they engage in a *homogeneous interaction*. If the agents belong to different algorithm families, their interaction is considered *heterogeneous*. The interaction between agents can also be characterized by the dataset they use: when two data agents operate on the same dataset or variations of it, this is known as a *unimodal interaction*; otherwise, its *multimodal interaction*.

Communication mechanism

Since under SoS-ML agent-agent interaction is defined as an exchange of information, the communication mechanism that facilitates these exchanges are a crucial component of the AI blueprint. The SoS-ML communication mechanism differentiates between *interaction platforms* and *information channels* and emphasizes the need for both. Interaction platforms collect evidence from interacting Data Agents, while information channels allow for straightforward information transfer, without any processing.

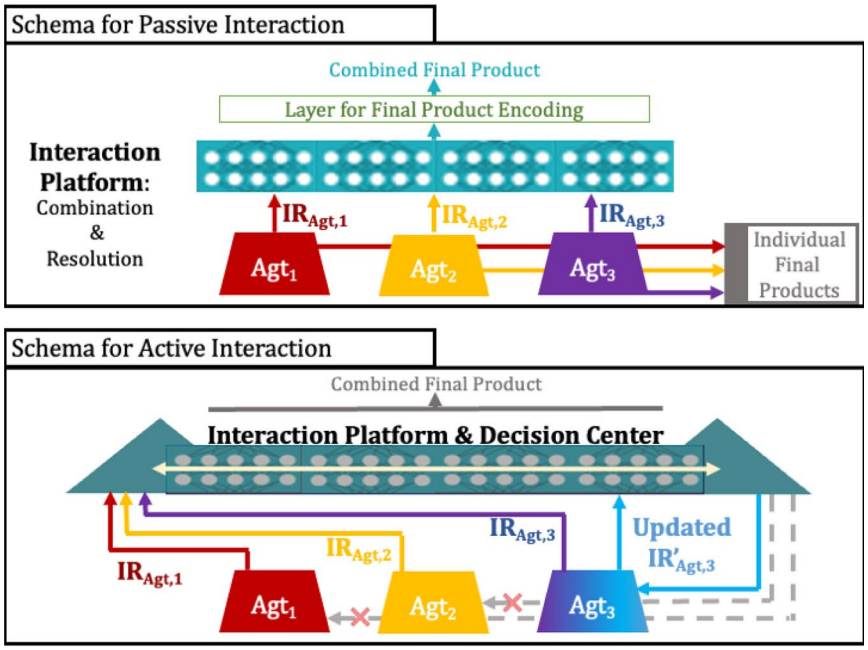
Interaction Platform. Interaction platforms collect evidence from interacting Data Agents. The Interaction Platform (Fig. 8) supports different types interactions by acting as a shared space where agents dock, exchange data, and collaborate.

Interaction Effect. The difference between the combined product and individual outputs is termed the interaction effect and can be calculated as follows (Fig. 8a):

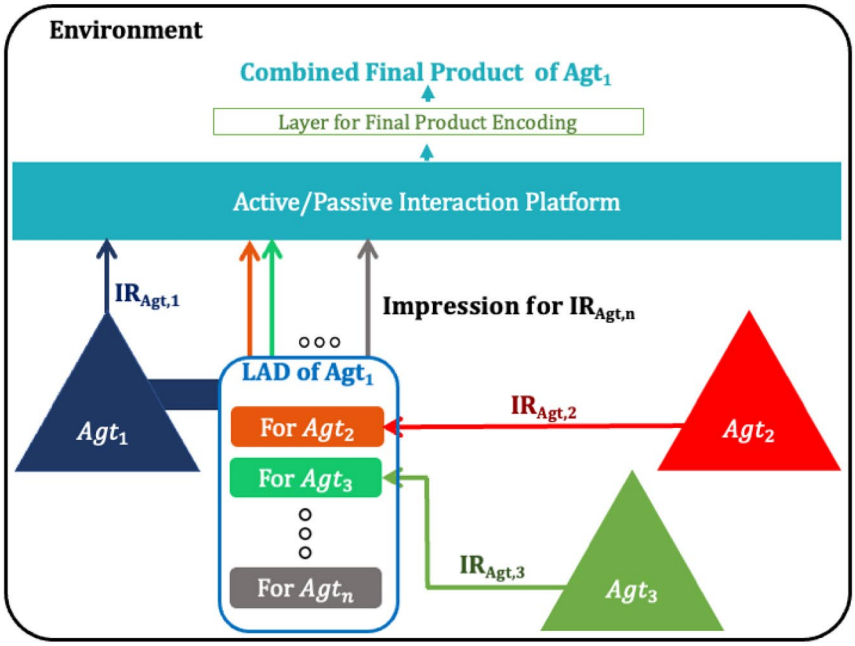
$$Interaction\ Effect = (CombinedFinalProduct) - (IndividualFinalProducts) \tag{8}$$

Interaction Platform Output. The output of an Interaction Platform is a decision-evidence pair. At this stage, this output pair may imbibe some contextual information. However, conversion of the decision-evidence pair to inference-evidence pair, is a task for the *supporting devices*.

Information Channels. These are passive form of information exchange or passage between System Agents to ensure that the system remains coordinated at all level.



(a) Environmental support for agents to interact in a passive or active manner. In the Active Interaction schema, the Decision Center would apply the selection pressure to determine which agent should be re-trained and obtain an updated internal representation.



(b) A figure depicting agent extension through the introduction of its own Language Acquisition Device (LAD) to interpret the knowledge gained by other agents. Here, only the Agt_1 is fully depicted; the other agents can also have their own LAD.

Fig. 8 Possible additional components of the environment to support agent-agent interactions

Supporting devices

The supporting devices within SoS-ML have three important functions: i) aid the System Agents convert the decision-evidence pair into inference-evidence pair, ii) work in tandem with System Agents to add or enhance contextual information to the inference-evidence pair, and iii) aid interaction platforms and information channels. Examples of supporting devices within the SoS-ML includes, LAD (excluding those acting as Agents or Interaction Platforms, Fig. 8b), and scaffolding and zones of proximity (Fig. 9).

Type of Inferences. Based on current literature [56, 57, 60], there are broadly four main types of inferences that can be used for information transfer between agents:

- Deduction: All $P(x)$ are $Q(x)$. 'a' was found/observed to be $P(x)$. 'a' is also $Q(x)$.
- Induction: 'a' is observed to be $P(x)$. 'a' is known to be $Q(x)$. Perhaps, all $P(x)$ is $Q(x)$.
- Abduction: 'a' is observed to be $Q(x)$. All $P(x)$ is $Q(x)$ is known. Perhaps, 'a' is also $P(x)$.
- Statistical: These inferences are more nuanced and difficult to generalize than the other three. Given a dataset, D, and a set of parameters, θ, statistical inferences can sometimes be written as maximizing the probability, $P(\theta|D)$.

Example of Inference-Evidence Pair. If the decision-evidence pair includes information such as "It is cloudy today" and "High humidity today", along with the system decision "Do not play tennis today" and the context "No information on indoor facilities", then the output might include inferences such as "Given clouds and high humidity, it will probably rain" (abduction) and "Based on past data, you do not like playing in the rain" (induction).

Obtaining explanations as a narrative

Since the SoS-ML supports the use of multiple ML-based models within an AI system, a large language model (LLM), serving as LAD, can be fine-tuned to take the inference-evidence pair and the system information from information channels as input, and produce an **explanation log**. The explanation log for a Phase 3 fXAI system is expected to have the following sections:

1. Overview for AI end-users: This outlines the purpose of the decision and presents the main inference, along with the supporting evidence for clarity and context.
2. System Information for AI Developers: This includes details on the contributing Data Agents, their recent updates, and specific focus areas. Additionally, it provides the Interaction Platform Output, which contains the classification result, supporting evidence,

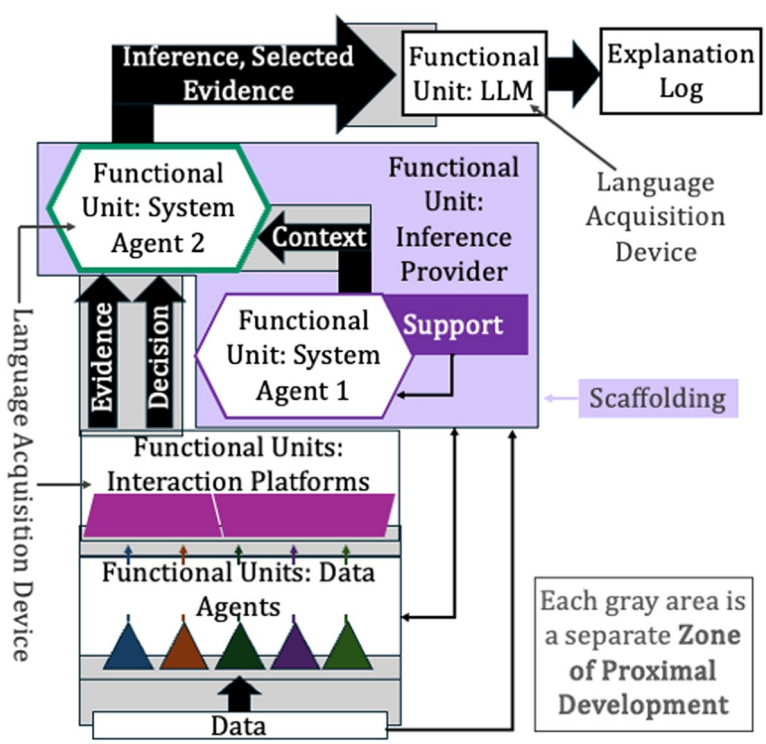


Fig. 9 SoS-ML guiding fXAI system architecture

and an uncertainty measure. The system also features other system-level information from System Agents, such as the context under which the decision was made.

3. Appendix for Interested Party: This provides detailed evidence logs and a system change log, ensuring transparency and enabling thorough tracking of updates the system made to itself.

$$Explanation\ Log = LLM(Contextual\ Explanation_{AI}) \tag{9}$$

In Phase 1, the explanation log is expected to produce evidence-based decisions, providing clear reasoning behind each decision made by the system. In Phase 2, the explanation log should additionally include information about the Data and System Agents, as well as some contextualized inference-evidence information, offering deeper insights into how the decisions were derived and the context in which they were made. Neither Phase 1 or Phase 2 is expected to produce narrative style explanation.

Summarizing the phases

The SoS-ML framework is designed to evolve through a structured, phased approach, where each phase builds upon the previous one to progressively enhance the system's capabilities. This phased development enables the system to begin with simple configurations and gradually incorporate more advanced functionalities such as system control, context detection, and scalable data processing. Each phase introduces specific improvements to agent collaboration, decision-making, and contextual awareness, ensuring that SoS-ML can adapt to increasingly complex tasks and provide robust, explainable AI solutions. Below, we outline the key characteristics and advancements within each phase of SoS-ML.

Phase 1: Multi-Agent Configuration. In Phase 1, SoS-ML operates with a basic multi-agent system where agents handle specific tasks or data subsets. Agents collaborate to improve prediction quality and increase confidence measures, without considering complex contextual factors.

Phase 2: System Control and Context Detection From Modularity. Phase 2 adds system control and initial context detection. Agents work in a structured environment, dynamically selecting optimal combinations based on task complexity, marking the first step toward context-aware decision-making arising from modularity.

Phase 3: Context Enhancement and System Scaling. Phase 3 enhances context awareness and scales up by adding more Data Agents for larger datasets and tasks. It improves information assimilation, decision-making, and produces explanation logs for transparency and accountability.

A possible fXAI system based on the SoS-ML blueprint

We conclude this "Methods: defining the architectural blueprint that fits the derived specifications" section, which discusses the SoS-ML blueprint, by providing an example of a potential functional XAI (fXAI) system constructed based on the SoS-ML framework.

Figure 9 presents the architecture of a functional XAI (fXAI) system, following the SoS-ML (System of Systems for Machine Learning) blueprint. The system is structured around multiple "Functional Units" that interact in layers. At the base are the Data Agents, which process raw data, passing information to Interaction Platforms that facilitate collaboration between agents. This is followed by two System Agents (System Agent 1 and System Agent 2) responsible for contextualizing and managing evidence. The Interaction Platform generates decisions and provide evidence to the Inference Provider (System Agent 2 and serving as a LAD), which supports higher-level inferences. The LLM (Large Language Model), acting as a LAD, helps produce an Explanation Log. The gray areas indicate Zones of Proximal Development (ZPD), supporting the interaction and growth of agents through scaffolding mechanisms.

Results and discussion

Proof of concept

Phase 1, Do multi-agent configuration lead to increased evidentiary information? Using Scenario 1, presented in "Methods: defining the architectural blueprint that fits the derived specifications" section, this study assessed whether interactions between various ML models can produce decision-making evidence and enhance the quality of extracted information, using the Pima Indian Diabetes dataset. To avoid confounding, we minimized external experimental factors and omitted the use of feature importance metrics such as the Gini impurity index. The algorithm selected included a k-nearest neighbor to represent traditional machine learning models prone to losing interpretability in higher dimensions, and a basic neural network with a single hidden layer of two neurons to exemplify more complex 'black-box' models. These models interacted via weighted majority voting, enabling direct evaluations of their combined impact on performance. Diagnostic features such as the number of pregnancies, and plasma glucose concentration, were used to predict diabetes onset. We introduced models, one at a time, with varying hyperparameters and feature subsets, incorporating bootstrapped samples to enhance diversity. This led to improved outcomes,

achieving an overall system accuracy of 80%, which is competitive with the usual 70% to 85% accuracy range seen in models such as Neural Networks on the Pima Indian dataset. Detailed analysis revealed that: i) the grouping [pedigree, BMI, glucose] showed a reliance of 81% with a 95% confidence interval of 75% to 87.5%, and a significant p-value; ii) the grouping [pregnancy, blood pressure] demonstrated a reliance of 76% within a confidence interval of 70% to 82%, also with a significant p-value; and iii) the grouping [skin thickness, insulin, age] had a 74% reliance but a non-significant confidence interval, suggesting less certainty in these variables. This demonstrates that model interactions enhance evidentiary information (e.g., reliance information) and improve information quality (e.g., confidence intervals), leading to more robust uncertainty or confidence measures compared to single black box models.

A Phase 2 fXAI system implementation

Implementing Scenario 2, described in "Methods: defining the architectural blueprint that fits the derived specifications" section, this study develops a Phase 2 fXAI system, the Pie Chart Interpreter, using deep machine learning agents structured into modules for pie chart analysis.

Phase 1 Development and Result. The first module, the Image Processing Functional Module, utilizes three agents-an Edge Processor, a Color Processor, and an Intensity Processor-to analyze and calculate the percentage values of pie slices based on color, edge, and intensity. Their predictions are then synthesized on an Interaction Platform to predict the slice sizes. The second module, the Numerical Relationship Identifier, employs agents to determine numerical relationships e.g. greater, smaller, or equal, linking quantitative data to symbolic representations. This deep model approach enables end-to-end training and exploration of numerical understanding in AI. Finally, a Translator module uses a fine-tuned Large Language Model to convert numerical and symbolic data into comprehensible English, enhancing data accessibility for users. The system's effectiveness is proven by high performance: the Image Processor achieves an L1-Norm percentage of 84.35%, the numerical relationship module records a mean accuracy of 99.54% and an AUROC of 99.95%, and user evaluations confirm the system's reliability (Fig. 10).

Phase 2 Development and Result. In the continued development of our pie chart fXAI system, the main goal is to intelligently select the optimal agent or combination of agents for interpreting pie charts, focusing on the number of slices as a critical contextual variable that indicates chart complexity. This experiment exclusively utilizes the Image Processing Functional Module, which consists of the Color Processor (denoted as Agent 1 or A1), the Intensity Processor (Agent 2 or A2), and the Edge Processor (Agent 3 or A3). A Q-Learning Agent, equipped with seven possible actions representing different combinations of these agents (A1, A2, A3, A1-A2, A1-A3, A2-A3, A1-A2-A3), applies an epsilon-greedy policy to explore and identify the most effective model combination. For example- a model combination of A2-A3 means that Agents 2 and 3 are chosen to process the pie chart image. During training, the agent selects a model combination for each pie chart, compares the output against the ground truth, and receives rewards based on the mean squared error (MSE) and additional metrics. These rewards help update its Q-table, refining its decision-making abilities. In the testing phase, the agent tests its strategies on new pie charts, recording all selected actions, predictions, and results for further analysis. The results are categorized by the number of pie slices, which helps reveal how the complexity of the chart influences model performance. For instance, the combination of the Color and Intensity Processors (Action A1-A2) is chosen 45% of the time for charts with three slices, achieving an average MSE of 0.1000. Additional data, such as the selection of the Intensity and Edge Processors (Action A2-A3) 88.46% of the time for charts with ten slices with an MSE of 7.99, suggests to experts that while intensity is the most stable feature for charts with fewer slices, the importance of edge information increases with the number of slices (Fig. 11).

Comparison of the Phase 2 Pie Chart Interpreter with the Phase 3 Architecture (Fig. 9). The Phase 2 Pie Chart Interpreter closely follows the architecture outlined in Fig. 9. However, unlike the Phase 3 architecture,

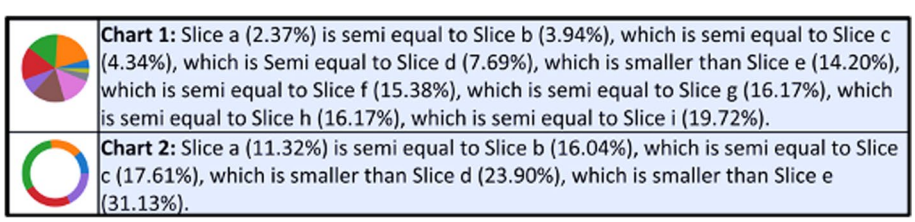


Fig. 10 Phase 1 output from the Pie Chart Interpreter

Category	3-5 Slices	5-7 Slices	7-10 Slices
Decomposition of Agent 2	97.37	95.05	83.78
Decomposition of Agent 3	96.72	94.08	86.68
Decomposition of Agent 1	98.03	96.13	92.38
Decomposition of Agents 1 and 3	98.57	96.51	93.57
Decomposition of Agents 1 and 2	44.13	58.94	63.33
Decomposition of Agents 2 and 3	96.37	95.88	85.33
No Decomposition	96.89	94.04	83.90

Fig. 11 Evidence from the decomposition experiment performed by the Phase 2 Pie Chart Interpreter to analyze pie charts

the Pie Chart Interpreter does not include System Agent 1, as additional contextual information is unnecessary for general pie chart interpretation. The Color, Edge, and Intensity Data Agents provide sufficient context for the Q-learning agent (System Agent 2 in Fig. 9)) to analyze and explain the charts. In more domain-specific applications, such as business or pharmaceuticals, System Agent 1 would be required to incorporate relevant context for more specialized interpretations.

Existing support for SoS-ML in literature review

This section examines how different AI systems and ML models in existing literature-such as AlphaGo, self-driving cars, ensemble methods, and GANs-align with or diverge from the SoS-ML framework. While some systems demonstrate multi-agent collaboration and continuous improvement, they often lack explicit modularity, focus on explainability, and adherence to responsible AI principles. This comparison highlights the strengths of SoS-ML in promoting transparency and ethical AI, compared to models that primarily focus on performance.

AlphaGo [61–63]. AlphaGo, built on multiple ML models, beat Go Grandmaster. Multiple deep networks were used to predict moves based on the current board state, while the Monte Carlo Tree Search model explored potential moves. Reinforcement learning enhanced performance through self-play. The decision-making subsystem made informed move selection based on the neural network and MCTS model. Operating on a dynamic loop, AlphaGo can refine strategies and achieve advanced gameplay continually.

AlphaGo fits the SoS-ML blueprint through its use of multiple ML models (neural networks and Monte Carlo Tree Search) acting as Data Agents, collaborating to predict moves and explore strategies. Its decision-making system, which selects moves based on these models, mirrors the role of System Agents in SoS-ML, while reinforcement learning and self-play reflect the continuous improvement process of SoS-ML systems. However, unlike SoS-ML, AlphaGo lacks explicit contextual agents (such as System Agent 1) that adapt outputs to specific contexts, as it focuses solely on the game of Go without requiring domain-specific interpretations. Additionally, SoS-ML emphasizes more flexible, interdisciplinary applications beyond a single task such as Go.

Self-Driving Car [64–66]. A self-driving car relies on a suite of ML paradigms to achieve autonomous driving. Convolutional Neural Networks work on object detection, while Recurrent Neural Networks process sequential data for trajectory prediction. Techniques such as Kalman filters integrate information from cameras, and LiDAR into a comprehensive perception. Decision-making models utilize reinforcement learning to translate decision outputs into precise vehicle commands. Feedback loops and adaptation ensure efficient and safe driving. However, self-driving cars was not created for the systematic development of a SoS for XAI. Also, autonomy of self-driving car leads to issues with law, ethics and human psychology.

A self-driving car aligns with the SoS-ML blueprint by employing multiple ML models (e.g., CNNs for object detection and RNNs for trajectory prediction) as Data Agents that collaborate to form a comprehensive perception of the environment. The reinforcement learning-based decision-making resembles the role of System Agents in SoS-ML, which translate inputs into actions. However, unlike SoS-ML, self-driving cars are not designed with explicit modularity or multi-agent collaboration aimed at explainability (XAI).

Ensemble Methods [67, 68]. Ensemble boosting involves sequential model-model interaction, with subsequent models learning from the mistakes of previous models through weighted updates. Stacking leverages hierarchical interaction, where lower-level models provide predictions as input features for a meta-model. Bagging combines predictions from diverse models,

benefiting from the indirect but collective model-model interaction. These ensemble methods thrive on strategically exchanging information, opinions, and insights among the models to enhance overall prediction accuracy.

Ensemble methods align with the SoS-ML blueprint through their use of model-model interaction. In boosting, models interact sequentially, with each new model learning from the errors of the previous ones, resembling SoS-ML's adaptive agent interactions. Stacking employs hierarchical collaboration, where lower-level models feed into a meta-model, similar to System Agents or scaffolding in SoS-ML combining insights. Bagging represents collective interaction, combining diverse models' outputs. However, ensemble methods primarily rely on model outputs or class labels for collaboration, whereas SoS-ML also enables the comparison of internal representations between models, allowing deeper insight and interaction beyond just outputs. SoS-ML further emphasizes explainability and modular design, while ensemble methods focus primarily on improving prediction accuracy.

Generative Adversarial Network [69, 70]. GAN is an interesting example of adversarial interaction where a generator and discriminator play a competitive game, and the process is controlled via the summation of loss functions. The generator generates synthetic data to fool the discriminator, while the discriminator aims to correctly distinguish between real and generated data, generating high-quality data.

GANs exemplify adversarial interaction, where two models-the generator and the discriminator-compete in a game-like setup. The generator creates synthetic data to deceive the discriminator, while the discriminator strives to distinguish between real and generated data. This interaction is controlled by the summation of loss functions, pushing both models to improve iteratively. Unlike SoS-ML, which emphasizes collaborative interaction between models or agents, GANs rely on competition to enhance the quality of data generation, focusing on adversarial learning rather than cooperative explainability or modular design.

Does SoS-ML conform to responsible AI?

The proposed SoS-ML blueprint, grounded in principles from human-centric sciences such as cognitive psychology, linguistics, and neuroscience, seeks to significantly improve AI systems by integrating modularity, transparency, and functional interpretability. By leveraging insights from these fields, SoS-ML enhances AI system performance across several critical dimensions: accuracy, generalizability, context awareness, and explainability.

Model accuracy

SoS-ML improves accuracy by refining how errors are managed across multiple specialized agents. Each agent processes a subset of the dataset, isolating errors associated with specific features or tasks. This modular approach allows for targeted refinements, where error-prone components can be adjusted independently, improving overall system accuracy. Different agents generate unique internal representations, which minimizes cascading errors across the system.

Moreover, increasing the number of models in SoS-ML introduces error diversity. While some models may share similar errors, others will make different errors, allowing their combined output to average out the individual mistakes. This principle, rooted in ensemble learning, ensures that as long as the models are diverse and their errors are uncorrelated, the system benefits from reduced error rates through mechanisms such as majority voting or weighted averages. However, for this error-compensation effect to be effective, it is crucial that models do not consistently share the same biases or errors, as this would limit the averaging effect.

Model accuracy, generalizability, downstream maintenance

SoS-ML enhances both accuracy and generalizability through its modular and multi-agent design. In terms of accuracy, each agent in SoS-ML is responsible for processing a specific subset of data or task, which localizes errors to those agents. This approach allows for targeted error corrections, where specific components are refined without impacting the overall system. By isolating errors, the risk of cascading issues is minimized, and the process of managing and maintaining the AI system becomes more streamlined over its lifecycle. This modular error correction also simplifies downstream system management, making it easier to identify and fix issues without overhauling the entire model.

SoS-ML also encourages developers to incorporate multiple models and datasets through phased development, enabling the system to average out errors across diverse models, similar to ensemble learning methods such as bagging. This ensures that different models, trained on different data views, can compensate for each other's errors, thus improving overall system accuracy.

In addition, generalizability improvements feed into accuracy. By incorporating multiple datasets and algorithms when possible, SoS-ML ensures that models can perform well across diverse contexts and unseen data, reducing overfitting. As agents generalize across varied scenarios, the system's predictions become more robust, enhancing accuracy. Moreover, System Agents can interact with humans, learning from feedback to adjust

predictions based on real-time behavior, further refining both generalizability and accuracy.

Thus, SoS-ML not only improves accuracy through modular error management but also facilitates easier lifecycle management of AI systems by isolating and correcting errors in specific components. Its phased development approach, which allows developers to integrate multiple models and datasets, enhances both generalizability and accuracy, making SoS-ML particularly well-suited for complex, real-world applications.

Context awareness

Traditional AI systems often lack contextual awareness, which can lead to errors in dynamic or unfamiliar environments (e.g., Tesla Autopilot's failure to distinguish a white truck from a bright sky). The SoS-ML framework enhances context awareness by incorporating System Agents that analyze various contextual factors, such as environmental variables, social conditions, and domain-specific factors. These agents ensure that predictions are informed by real-world nuances, making them more reliable in complex situations.

For instance, in autonomous driving, context-aware agents can adjust their focus dynamically based on changing conditions such as lighting, road types, or traffic patterns, leading to more accurate and safer decision-making. By continuously adapting to context, SoS-ML mitigates static, context-insensitive predictions, making it more suitable for real-world applications.

Explainability

SoS-ML significantly enhances explainability, by focusing on two key aspects: evidence accuracy and inference accuracy.

Evidence Accuracy. SoS-ML derives explanations directly from the internal model representations, allowing users to trace how specific agents processed data. This transparency ensures that explanations are grounded in actual features or data that contributed to the decision, with clear confidence levels. By providing evidence at the component level, SoS-ML reduces the risk of incomplete or biased explanations, which are common in post-hoc methods such as SHAP or LIME.

Inference Accuracy. SoS-ML goes beyond justifying individual predictions by enabling agents to synthesize explanations based on context and functional interpretations of data. This means the system can explain not just "what" a decision was, but "why" it was made in a specific context. For instance, in healthcare, SoS-ML would explain how lab results were weighted alongside patient history, ensuring higher inference accuracy. This approach provides users with transparent, traceable evidence and contextual explanations that reflect the underlying decision-making process, improving both the clarity and reliability of AI-generated decisions.

Identifying New Evaluation Metrics SoS-ML's focus on modularity and internal representations enables the identification of new areas where evaluation metrics are needed. For example:

- Metrics could be developed to assess the effectiveness of agent collaboration, measuring how well individual agent outputs are combined into a unified decision.
- Context-specific evaluation metrics could be created to measure how well context-aware agents adjust predictions based on dynamic real-world conditions.
- New metrics for modular error tracing could assess how isolated each agent's errors are and how much they influence the overall system's accuracy, leading to more targeted improvements.

End-to-end training vs. modular approach in SoS-ML

SoS-ML emphasizes a modular architecture that allows for end-to-end training within specific functional units but discourages full end-to-end training of the entire AI system. This contrasts with approaches in the development of AlphaZero, which is seen as an upgraded, fully end-to-end version of AlphaGo (even though AlphaGo defeated the human Go Master). While the computer science research community often prioritizes developing end-to-end models for their efficiency and performance gains, such an approach poses challenges when applied to socially critical systems.

In particular, full end-to-end models require a mathematical description of not only technical performance but also the social issues embedded in these systems, such as fairness, transparency, and accountability-beyond just explainability. This creates a trade-off for AI developers: should the focus be on optimizing system performance through end-to-end training, or should social inclusivity and explainability take precedence? As the landscape of AI development evolves, this decision will depend on the specific goals of the system. While not all AI systems require explainability, those in high-stakes domains such as healthcare, finance, or law may benefit from approaches that prioritize social factors alongside technical considerations.

Conclusion

In this paper, we introduced System of Systems for Machine Learning (SoS-ML) as a novel framework for addressing key challenges in explainable AI (XAI). Through experiments on the Pima Indian Diabetes dataset and the Pie Chart Interpreter, we demonstrated how

SoS-ML enhances explanation accuracy, context-awareness, and generalizability by integrating insights from social and cognitive sciences. Our results showed that SoS-ML's modular, multi-agent approach improves both model accuracy and interpretability, while its ability to incorporate human feedback contributes to more adaptable and socially responsible AI systems.

The theoretical implications of this work lie in its potential to reshape how XAI systems are designed, moving beyond post-hoc explanations towards functional interpretability and evidence-based explanations. By aligning with theories like Vygotsky's Social Interactionist Theory and Chomsky's Innatist Theory, SoS-ML positions itself as a bridge between computational and social sciences, promoting collaboration among models to enhance AI's transparency and accountability.

Practically, the application of SoS-ML in high-stakes environments such as healthcare and autonomous systems demonstrates its potential to improve decision-making, safety, and user trust. By embedding explanation mechanisms directly into the system, SoS-ML offers more reliable and contextual explanations, reducing risks associated with traditional black-box models. Additionally, the ability to continually learn from both new data and human interactions highlights the system's robustness in dynamic and evolving environments.

However, some limitations of this study should be noted. First, while the modular approach promotes flexibility, its computational complexity could limit scalability in resource-constrained environments. Second, the full potential of human-agent interaction requires further exploration, particularly in terms of how biases from human feedback could influence system behavior. Finally, additional evaluation metrics are needed to fully assess how well SoS-ML conforms to the social specifications outlined in this paper.

In conclusion, while challenges remain, SoS-ML offers a promising path forward for XAI, emphasizing modularity, contextual awareness, and explainability at the core of AI development. By aligning AI system design with principles from social sciences, SoS-ML has the potential to make AI more responsible, transparent, and ethically aligned with real-world needs.

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Authors' contributions
The idea is presented and written by RS. This paper uses evidence (presented in "Results and discussion" section) that was produced from experimentation conducted by PG (Master Thesis) and HT (Ph.D., pending paper in IEEE Transactions on Pattern Analysis and Machine Intelligence). RS supervised both PG and HT.

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